

Chua Oscillator: Experiment Plan

Summer 2026 — v5

This plan accompanies the lab guide. The guide sets out the concepts and the pre-lab calculation; this document says what we actually do at the bench, in what order, and what is optional.

The programme is split into **must-have** and *nice-to-have*. The must-have part is designed to be reachable even though the potentiometer dial is coarse and cannot be set blind to a target value. Anything that needs specific resistances on an even grid has been moved to nice-to-have.

The setup

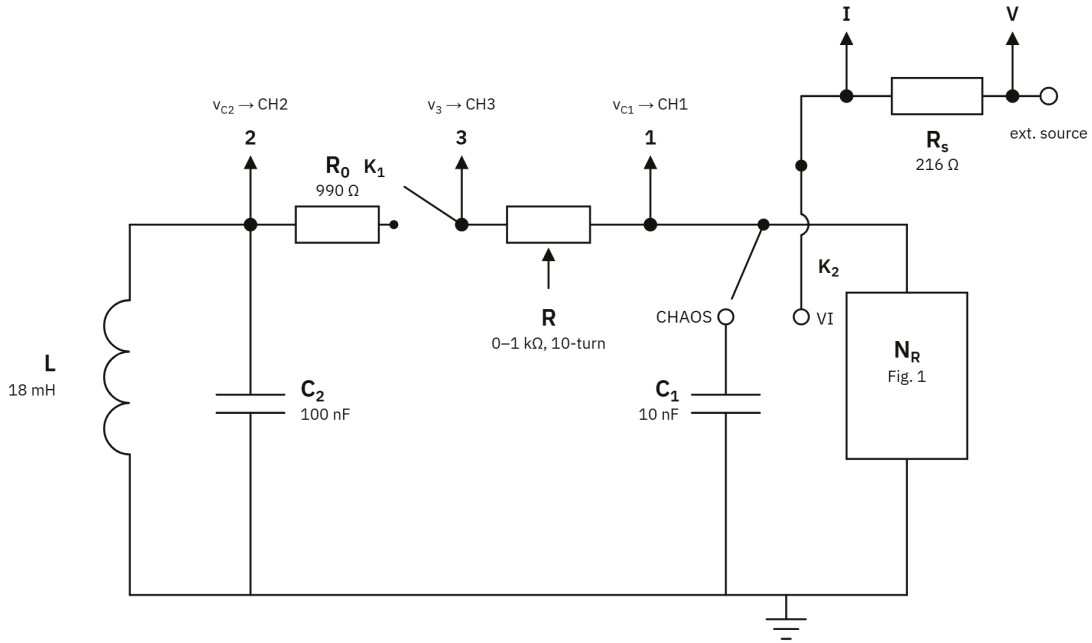


Figure 1: The oscillator as built. Resonant part L , C_2 ; coupling resistance $R_0 + R$ with disconnect switch K_1 ; C_1 ; and the nonlinear element N_R . Probe points: 1 = V_1 (CH1), 2 = V_2 (CH2), and 3 = V_3 , the node between R_0 and R (CH3). K_2 selects oscillator mode (CHAOS) or the V – I measurement of N_R (VI, with K_1 open), for which a triangular sweep is applied at terminal V and the current is taken across the shunt R_s .

Values: $L = 18\text{ mH}$, $C_2 = 100\text{ nF}$, $C_1 = 10\text{ nF}$, $R_0 = 990\ \Omega$ measured, R a $0\text{--}1\text{ k}\Omega$ ten-turn potentiometer, shunt $R_s = 216\ \Omega$. Oscilloscope: Agilent InfiniiVision DSO-X 3014A.

The key simplification: reading R without opening the circuit

Opening K_1 to measure R with a multimeter kills the oscillation and resets the state — and the state is not determined by R alone. Instead we use node 3.

No current flows into a scope probe, so node 3 is a plain voltage divider between V_1 and V_2 . Since R_0 sits on the V_2 side and the potentiometer on the V_1 side,

$$V_3 = \frac{R_0}{R_0 + R} V_1 + \frac{R}{R_0 + R} V_2,$$

so fitting $V_3 = AV_1 + BV_2 + C$ over a whole record gives

$$R = R_0 \frac{B}{A}$$

The fitted constant C absorbs any channel offset. Run it as `python rpot.py record.csv`; the printed fit residual is the health check (about 1 % is normal, above 5 % means a probe has moved or a channel is clipping).

Consequence for the procedure: we never “set R to $1750\ \Omega$ and record”. We turn the knob slowly, watch the scope, record when something worth recording appears, and find out afterwards what R was.

Absolute accuracy is a few percent, dominated by the relative gain of the channels; relative accuracy across a sweep is a fraction of an ohm. That is the right way round — a constant offset shifts the whole picture without changing any spacing.

Must have

M1. V – I characteristic of N_R

K_1 open, K_2 in position VI. Apply a slow triangular sweep at the outer terminal V, 20 Hz to 100 Hz, with amplitude enough to pass ± 7 V at the element so that both saturation segments are captured. Then

$$v = V_I, \quad i = \frac{V_V - V_I}{R_s}.$$

Check that the forward and backward sweeps overlap; if they open into a loop, lower the frequency.

Fit straight segments: G_a (inner), G_b (outer, left and right separately — they are not equal), G_c (saturation), and the breakpoints.

Two traps: set the segment boundaries at the breakpoints you find rather than by eye, and check the intercept of the inner segment — a nonzero value means the whole curve is offset, which barely moves the slopes but directly moves the equilibrium positions.

M2. Predicted range for three equilibria

From the fitted slopes alone, before touching the oscillator:

$$\frac{1}{|G_a|} < R_{\text{total}} < \frac{1}{|G_b|}, \quad R_{\text{total}} = R_0 + R.$$

Three equilibria are necessary for a double scroll, so this bounds where it can live. Convert to dial readings and write the prediction down *before* looking. This is the same calculation as in the pre-lab, now with the students’ own measured slopes instead of the canonical ones.

M3. Find the double scroll

K_2 in position CHAOS, K_1 closed. Turn the knob slowly and watch the XY display of CH2 against CH1 (V_2 vertical, V_1 horizontal). Record whenever the picture changes character, and in particular at the two resistances where the second lobe appears and disappears. Records of 200 ms at 20 ms per division; each holds about 570 cycles.

Then run `rpot.py` over the files and sort by R . Compare the observed double-scroll range with M2: it must lie inside.

Take at least three long, clean records well inside the range — these carry the rest of the analysis.

M4. Lorenz map

From one long double-scroll record: smooth V_1 lightly (Savitzky–Golay, window about 1/20 of a period), find the local maxima with a prominence threshold of 2% of the range, and plot M_{n+1} against M_n . Draw the diagonal.

If the points fall on a curve rather than a cloud, the three-dimensional flow has reduced to a one-dimensional map — the experimental signature of a horseshoe, and the strongest single result in the experiment. Repeat for two or three records at different R and show how the map changes.

M5. Lyapunov exponent

From the same map: estimate the local slope $f'(M_n)$ by a sliding linear fit along M_n , and combine with the mean return time,

$$\lambda \simeq \frac{\langle \ln |f'(M_n)| \rangle}{\langle T \rangle}.$$

Two traps worth stating in the report. Use the *mean* return time, not the median — the distribution is skewed because lobe switches take about twice as long as an ordinary winding, and the difference exceeds 10%. And average $\ln |f'|$ over the points actually visited, not from one straight-line fit per branch: the neighbourhood of the turning point, where the slope drops toward zero, pulls the average down substantially.

Quote it as an estimate from the return map, not a direct measurement; a proper Rosenstein or Wolf calculation from the time series typically comes out 10% to 20% lower.

Nice to have

Only if the must-have programme is finished and working.

N1. The Hopf point.

The value of R at which the outer equilibria stop being resting states and an oscillation is born, i.e. where the real part σ of the complex eigenvalue pair changes sign:

$$R_{\text{Hopf}} = \frac{-G_b(1 + C_1/C_2)}{G_b^2 + C_1^2/(C_2L)}.$$

Measure the onset of oscillation and compare, and compare the frequency there as well. Cheap, and the agreement is usually striking — the best first choice if any time is left.

N2. Phase-portrait gallery.

Limit cycle, period-2, single scroll, double scroll: four clean records for the report.

N3. Bifurcation diagram.

Local maxima of V_1 against R , one column per record. With a coarse dial this will be unevenly sampled; present it as a scatter of the points you have, not as a uniform sweep.

N4. Two directions and hysteresis.

Sweep up and down without opening the circuit and overlay. Where the branches differ there are coexisting attractors. Be honest if the sampling is too sparse to resolve it.

N5. Period doubling and Feigenbaum.

Needs a fine, even step — the hardest item on this list with a coarse dial.

N6. Shilnikov ratios.

Compute the eigenvalues at all three equilibria from your own measured slopes and check $|\sigma| < |\gamma|$.

Oscilloscope settings — check at the start of every session

1. **Channel mapping.** CH1 = V_1 (node 1), CH2 = V_2 (node 2), CH3 = V_3 (node 3), as labelled in Figure 1. These are the defaults assumed by `rpot.py`. Verify once at the start: V_1 is the one with the larger swing, several volts, while V_2 stays below a volt. Swapping CH1 and CH2 makes `rpot.py` return a resistance outside the range of the potentiometer, which is the quickest way to notice.
2. **DC coupling on all three channels.** AC coupling deletes exactly the information that distinguishes a single scroll from a symmetric orbit.
3. **20 MHz bandwidth limit ON.** The signal is 2.85 kHz; the filter is free.
4. **Fix V/div once and leave it.** Changing a range changes that channel's gain by a percent or two, which is tens of ohms in the divider result.
5. **Probe attenuation in the menu must match the probe switch.**
6. **$\times 10$ probes, compensated** against the Probe Comp square wave. $\times 1$ probes add about 90 pF to node 1 — 0.9% of C_1 , enough to shift boundaries by a few ohms.
7. **Acquisition Normal**, not Peak Detect or Average.
8. **20 ms per division**, giving a 200 ms record.

Time budget

Three measurement meetings of six hours.

Meeting A.

Setup and channel check, V – I curve (M1), fit, and the M2 prediction. Then a first hunt for the double scroll. Must end with the fitted curve and a written prediction.

Meeting B.

M3 in earnest: both boundaries of the double-scroll range, plus the long records. Start M4 on the spot, so you know whether the records are good enough before the equipment goes away.

Meeting C.

Finish M4 and M5. Remaining time goes to the nice-to-have list, starting with N1 and N2.

Two things that can still ruin it

1. **A wrong channel mapping, AC coupling, or a changed V/div.** Each quietly corrupts either the resistance scale or the asymmetric regimes.
2. **Records that are too short.** A few cycles cannot show a lobe switch, so the attractor looks single-scrolled even when it is not. 200 ms at 20 ms per division.